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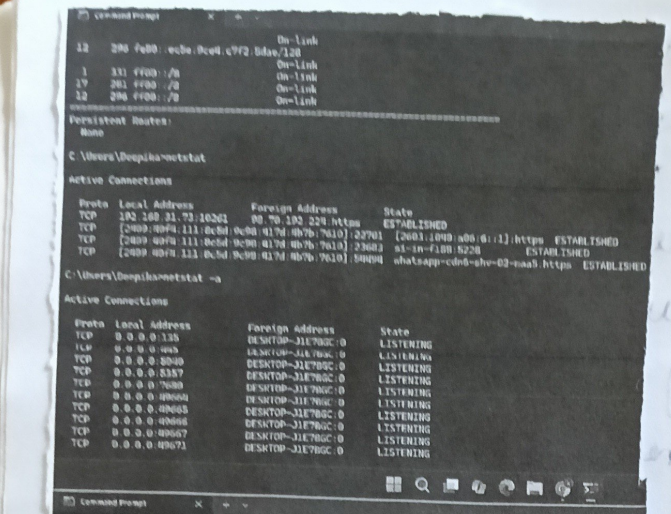
Basic networking commands in Linux and windows system

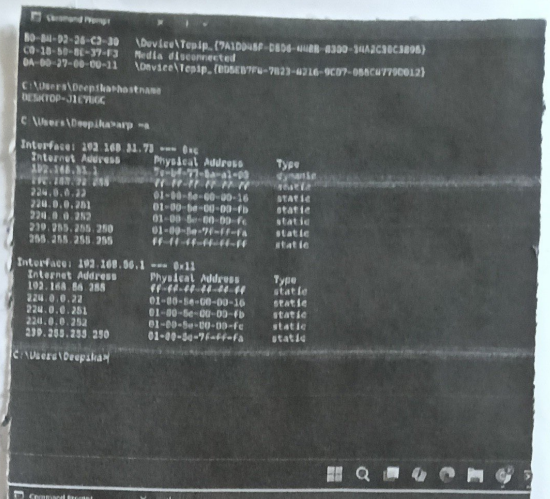
Aim: To study and execute basic networking commands in windows and Linux operating systems for viewing network configuration, testing connectivity checking routing information monitoring network status and troubleshooting network related issues

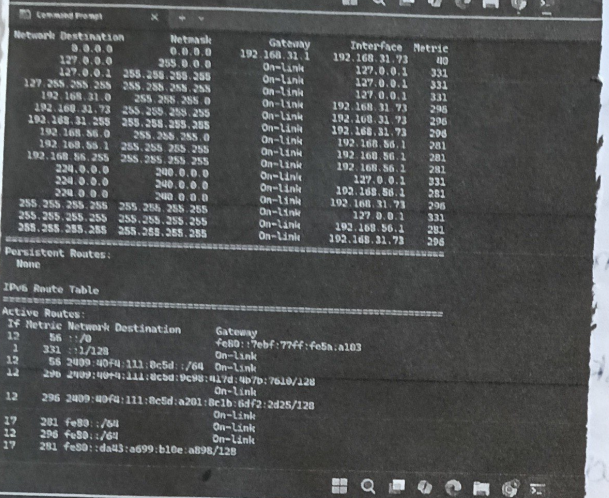
Requirements: window 10/11 or Linux (ubuntu 20.04/22.04)
- command prompt/windows
- Linux
- Active network (LAN/wifi)

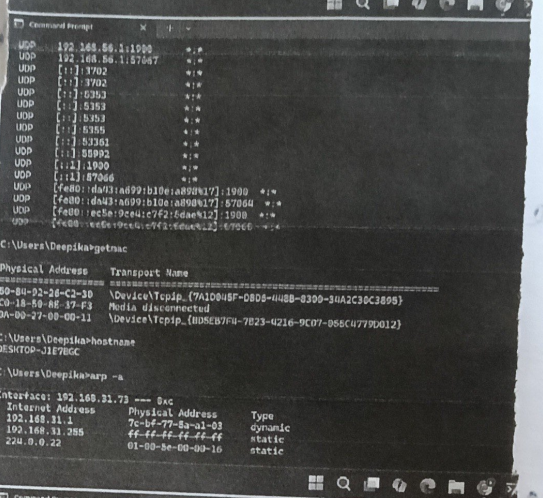
- ip-config / ifconfig / IP - displays network configuration
- ping - tests connectivity to a remote host
- tracert / traceroute - Displays the path packets take to reach a destination
- nslookup - resolves domain name to IP address
- netstat - Displays active network connections ports
- arp - displays ARP cache.
- hostname - displays the system hostname
- route - displays or modifies the routing table.

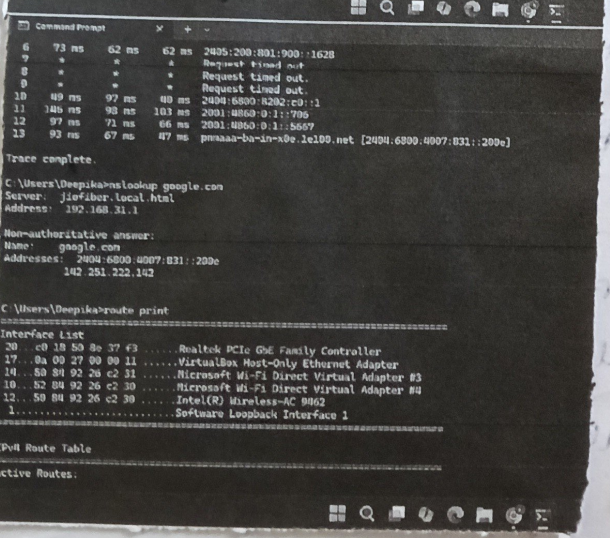
→ windows commands:
- commands: ipconfig
  displays: IP address
  subnet mask
  Default gateway

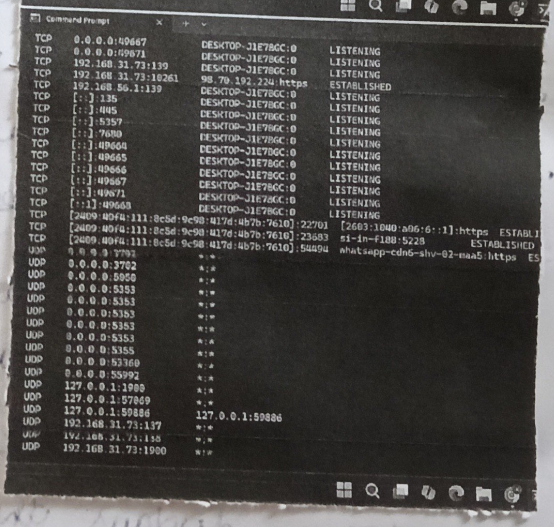


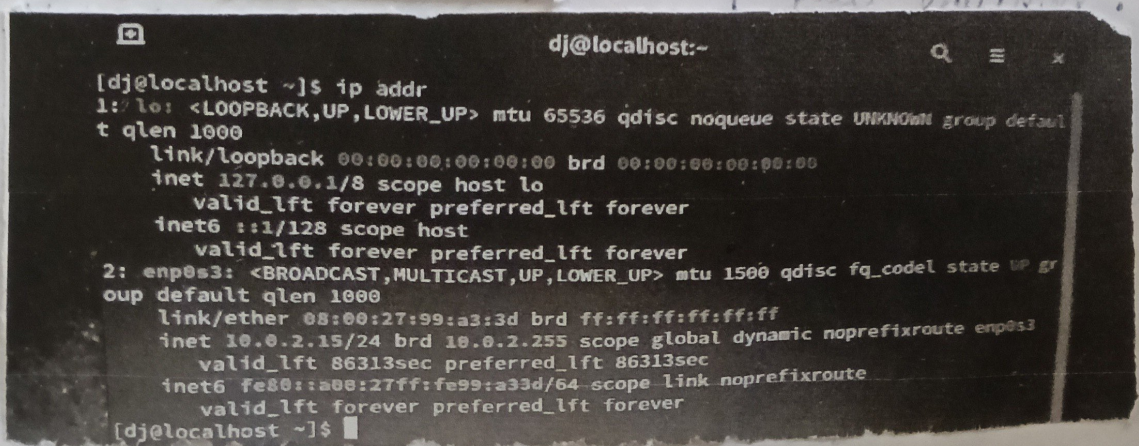


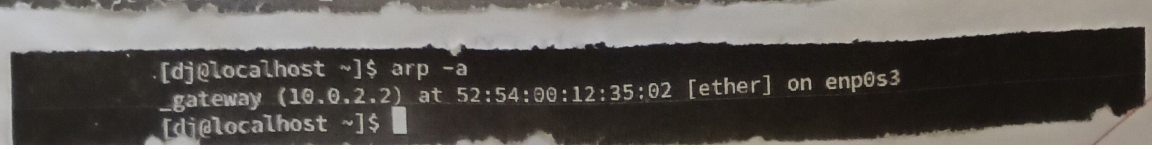









dj@localhost:~
[dj@localhost ~]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
link/ether 08:00:27:99:a3:3d brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
valid_lft 86313sec preferred_lft 86313sec
inet6 fe80::a00:27ff:fe99:a33d/64 scope link noprefixroute
valid_lft forever preferred_lft forever
[dj@localhost ~]$


[dj@localhost ~]$ arp -a
_gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enp0s3
[dj@localhost ~]$

Real information

- command : ping google.com
  Display : checks whether the destination is reachable

- command : tracert google.com
  purpose : displays every router (hop) between your computer and destination

- command : nslookup google.com
  displays : DNS server
  IP address of website

- command : route print
  Displays : network route
  Gateway
  Destination networks

- commands : netstat
  displays : Active TCP connections
  Listening ports

- command : netstat -a
  displays : all port
  TCP connections
  UDP ports

- command : getmac
  purpose : displays MAC address of the computer

- command : hostname
  purpose : displays computer name

- command : arp -a
  purpose : displays IP-to MAC address mappings

```
[dj@localhost ~]$ netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:ipp           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:ssh             0.0.0.0:*               LISTEN
tcp6       0      0 localhost:ipp           [::]:*                  LISTEN
tcp6       0      0 [::]:ssh                [::]:*                  LISTEN
udp        0      0 localhost.locald:bootpc _gateway:bootps         ESTABLISHED
udp        0      0 0.0.0.0:mdns            0.0.0.0:*
udp        0      0 localhost:323           0.0.0.0:*
udp6       0      0 [::]:mdns               [::]:*
udp6       0      0 localhost:323           [::]:*
raw6       0      0 [::]:ipv6-icmp          [::]:*                  7
Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  2      [ ACC ]     STREAM     LISTENING     33115    /run/user/0/gnupg/d.
w35a3wcth45h7ype6ibtf71/S.gpg-agent
unix  2      [ ACC ]     STREAM     LISTENING     20832    /run/systemd/private
unix  2      [ ACC ]     STREAM     LISTENING     33116    /run/user/0/gnupg/d.
w35a3wcth45h7ype6ibtf71/S.gpg-agent.extra
unix  2      [ ACC ]     STREAM     LISTENING     33117    /run/user/0/gnupg/d.
w35a3wcth45h7ype6ibtf71/S.gpg-agent.browser
unix  2      [ ACC ]     STREAM     LISTENING     33118    /run/user/0/gnupg/d.
w35a3wcth45h7ype6ibtf71/S.gpg-agent.ssh
```

```
unix  3      [ ]         STREAM     CONNECTED     31720    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     32189    @480bc
unix  3      [ ]         DGRAM      CONNECTED     29346
unix  3      [ ]         STREAM     CONNECTED     32916
unix  3      [ ]         STREAM     CONNECTED     30206    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     36022
unix  3      [ ]         STREAM     CONNECTED     26939
unix  3      [ ]         STREAM     CONNECTED     23768
unix  3      [ ]         STREAM     CONNECTED     32188
unix  3      [ ]         STREAM     CONNECTED     31522
unix  3      [ ]         STREAM     CONNECTED     29431
unix  3      [ ]         STREAM     CONNECTED     30861
unix  3      [ ]         STREAM     CONNECTED     22675
unix  2      [ ]         DGRAM      CONNECTED     22313
unix  3      [ ]         STREAM     CONNECTED     32549    @480bc
unix  3      [ ]         STREAM     CONNECTED     36023    /run/dbus/system_bus_
socket
unix  2      [ ]         DGRAM                    28785
unix  3      [ ]         STREAM     CONNECTED     32470    /run/systemd/journal/
stdout
unix  3      [ ]         STREAM     CONNECTED     30187    /run/user/1000/pipewi
re-0-manager
unix  3      [ ]         STREAM     CONNECTED     29806    /run/systemd/journal/
stdout
```

```
[dj@localhost ~]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
    link/ether 08:00:27:99:a3:3d brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 86313sec preferred_lft 86313sec
    inet6 fe80::a00:27ff:fe99:a33d/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

```
[dj@localhost ~]$ hostname
localhost.localdomain
[dj@localhost ~]$ ping google.com
PING google.com (142.251.222.174) 56(84) bytes of data.
64 bytes from pnmaaa-an-in-f14.1e100.net (142.251.222.174): icmp_seq=1 ttl=113 t
ime=137 ms
64 bytes from pnmaaa-an-in-f14.1e100.net (142.251.222.174): icmp_seq=2 ttl=113 t
ime=171 ms
64 bytes from pnmaaa-an-in-f14.1e100.net (142.251.222.174): icmp_seq=3 ttl=113 t
ime=57.3 ms
64 bytes from pnmaaa-an-in-f14.1e100.net (142.251.222.174): icmp_seq=5 ttl=113 t
ime=358 ms
64 bytes from pnmaaa-an-in-f14.1e100.net (142.251.222.174): icmp_seq=8 ttl=113 t
ime=445 ms
^C
--- google.com ping statistics ---
9 packets transmitted, 5 received, 44.4444% packet loss, time 8109ms
rtt min/avg/max/mdev = 57.289/233.532/444.882/144.424 ms
[dj@localhost ~]$
```

```
[dj@localhost ~]$ nslookup google.com
Server:         192.168.230.203
Address:        192.168.230.203#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.222.174
Name:   google.com
Address: 2404:6800:4007:83b::200e
```

```
[dj@localhost ~]$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::a00:27ff:fe99:a33d  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:99:a3:3d  txqueuelen 1000  (Ethernet)
        RX packets 790095  bytes 1146907873 (1.0 GiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 80488  bytes 4889352 (4.6 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

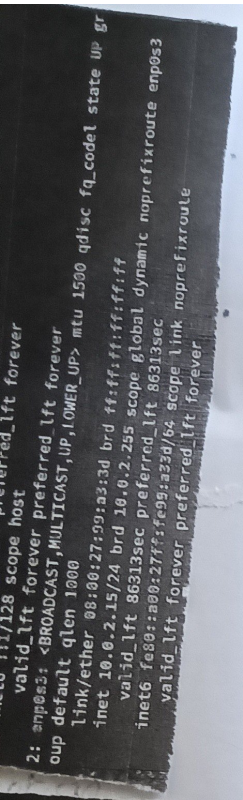
lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 17  bytes 2062 (2.0 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 17  bytes 2062 (2.0 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

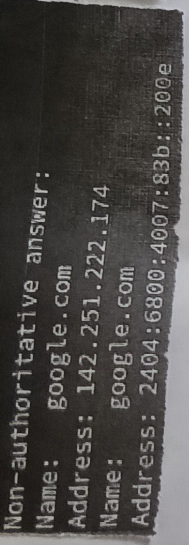
[dj@localhost ~]$
```

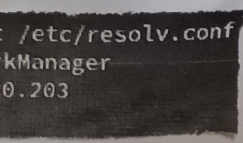
```
[dj@localhost ~]$ cat /etc/resolv.conf
# Generated by NetworkManager
nameserver 192.168.230.203
[dj@localhost ~]$
```

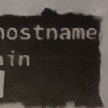
```
[dj@localhost ~]$ hostname
localhost.localdomain
[dj@localhost ~]$
```

```
[dj@localhost ~]$ curl google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
[dj@localhost ~]$
```









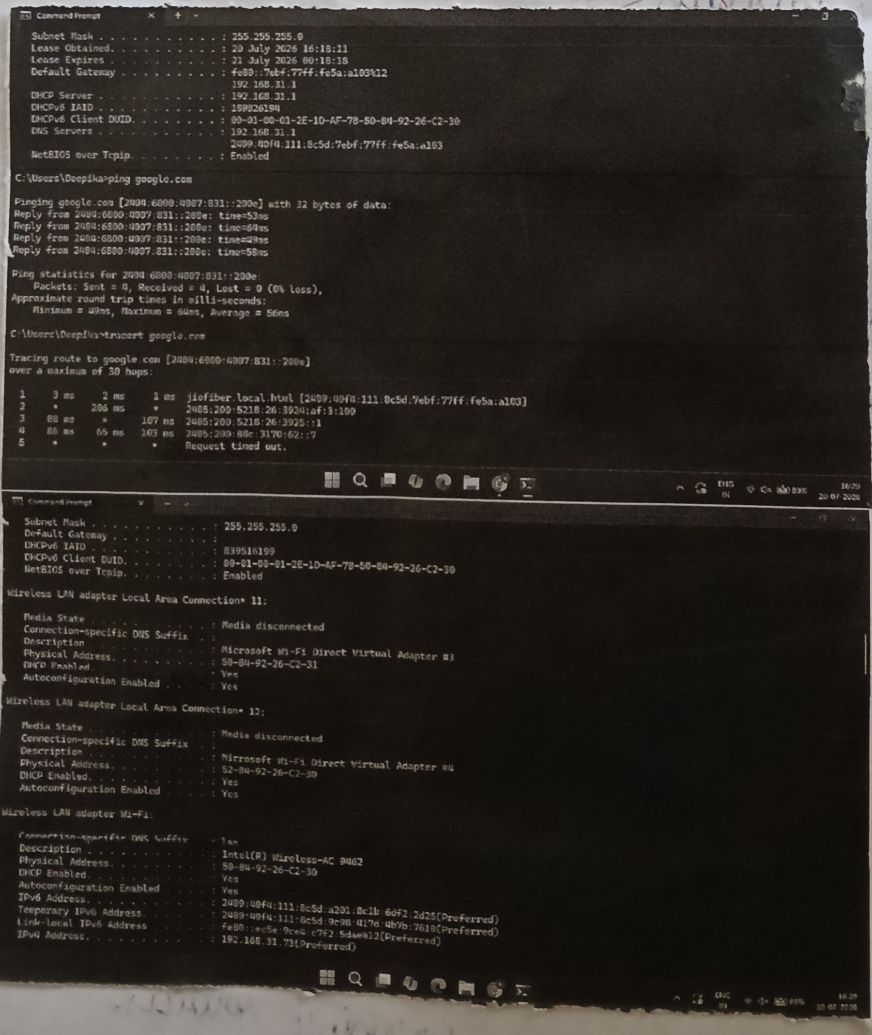
→ Linux networking commands:

- command: ip address or ipconfig
  - purpose: displays IP address
- command: ip route
  - purpose: shows: default gateway
    - Routing table
- command: hostname
  - purpose: displays system hostname
- command: ping google.com
- command: traceroute google.com
  (if not installed :)
  sudo apt install traceroute
- command: netstat -a(o) ss -a
- purpose: displays network statistics
- command: arp -a(oi) ip neigh
- purpose: displays ARP table
- command: ip link
- purpose: displays MAC address
- command: cat /etc/resolv.conf
  - displays: DNS configuration
- command: ip link show
- displays: UP / DOWN status
  - interface name
- purpose: check interface status
- command: ss -tuln
- purpose: Displays: TCP ports
  - UDP ports
  - listening services.

```
[dj@localhost ~]$ ss -tuln
Netid    State     Recv-Q    Send-Q        Local Address:Port        Peer Address:Port
udp      UNCONN    0         0               0.0.0.0:5353             0.0.0.0:*
udp      UNCONN    0         0             127.0.0.1:323              0.0.0.0:*
udp      UNCONN    0         0                 [::]:5353                [::]:*
udp      UNCONN    0         0                [::1]:323                 [::]:*
tcp      LISTEN    0         4096          127.0.0.1:631              0.0.0.0:*
tcp      LISTEN    0         128             0.0.0.0:22               0.0.0.0:*
tcp      LISTEN    0         4096             [::1]:631                 [::]:*
tcp      LISTEN    0         128              [::]:22                   [::]:*
[dj@localhost ~]$
```

```
[dj@localhost ~]$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
[dj@localhost ~]$
```

```
[dj@localhost ~]$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT
group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mo
de DEFAULT group default qlen 1000
    link/ether 08:00:27:89:a3:3d brd ff:ff:ff:ff:ff:ff
[dj@localhost ~]$
```

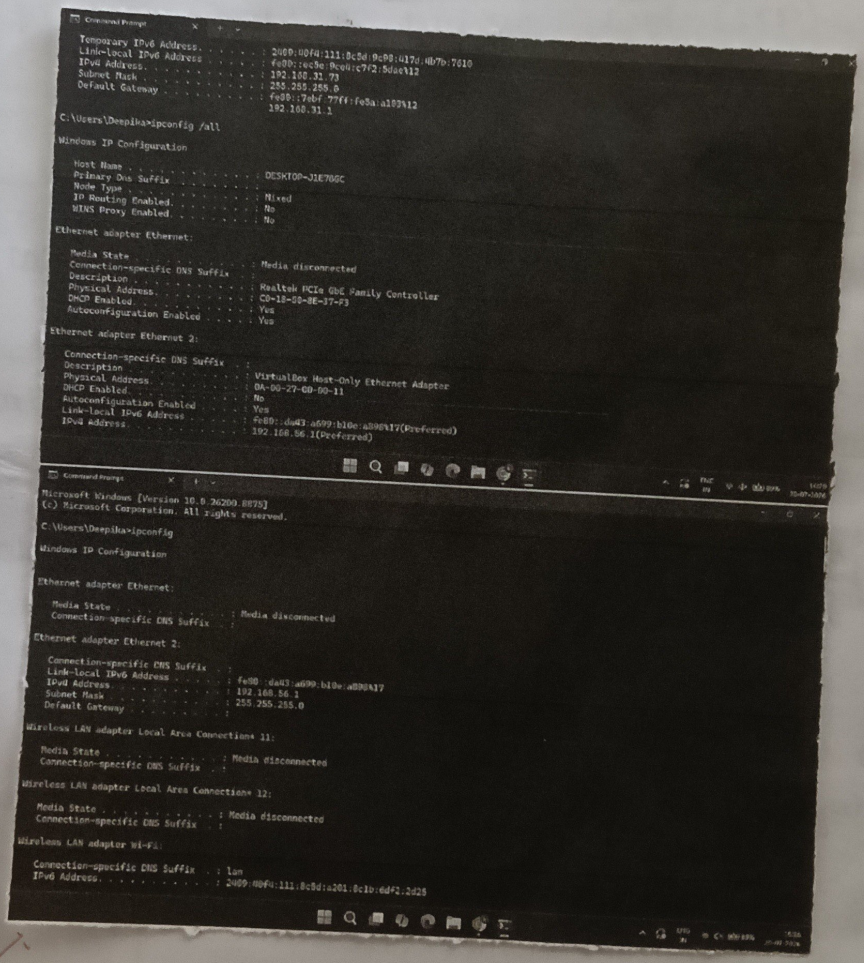


• command
• purpose
• command
• purpose

Result:

this windows

- command : arp -a
- purpose : displays IP-to-MAC address mappings
- command : netstat -a
- purpose : All ports
  TCP connections
  UDP ports.



Result:

Thus the commands of Linux and windows are successfully run.